

Making Accessible PDFs with L^AT_EX

In this guide, I will go over some important things to know to get started making accessible PDFs with LaTeX. Much of this was taken from [the LaTeX tagging project](#). It should be noted that LaTeX documents take significantly longer to compile when making them accessible.

Initial Setup

First, it is important that your version of LaTeX is updated to the **2025 version** or newer. You can install the newest version from the LaTeX website: <https://www.latex-project.org/get>. Once you have installed the 2025 version, **make sure all the packages are updated** by opening TeX Live or TeX Live Utility application.

Metadata

At the beginning of each LaTeX document, before the `\documentclass` statement, you need to put the `\DocumentMetadata` statement. This is how LaTeX knows some of that important accessibility information. Here we can set the language, indicate the type of PDF to create, and set up the tagging. Here is an example of a statement that works well with the Ally accessibility checker on Canvas:

```
\DocumentMetadata{
  lang=en-US,      % It is also fine to put lang=en
  pdfversion=1.7,
  pdfstandard=ua-1,
  tagging=on
}
```

While PDF version 2.0 with standard ua-2 are more modern, the Ally checker does not work with PDF 2.0.

If the `tagging=on` causes issues, you can put `testphase={phase-III,math,table,title}` instead. This has similar functionality to `tagging=on`, but can work with 2023 or 2024 versions of LaTeX.

PDF Engine

It is recommended to use the LuaLaTeX engine rather than pdfLaTeX, but either can work. XeLaTeX is not compatible. If using LuaLaTeX, then the line `tagging-setup={math/setup=mathml-SE}` can be added in `\DocumentMetadata` to create a separate MathML html file that contains all the mathematics expressions and is supposedly readable by screen readers. However, I don't think this file can currently be used with Ally.

The engine that LaTeX uses to create a specific PDF can be set with a magic comment as the first line of the document, just before the `\DocumentMetadata` command. Magic comments:

```
% !TeX program = lualatex   will render in LuaLaTeX.
% !TeX program = pdfflatex  will render in pdfLaTeX.
```

When using the LuaLaTeX engine, it is recommended to use the `unicode-math` package by adding `\usepackage{unicode-math}` since that exports to MathML much better. This conflicts with the `amssymb` and `amsmath` packages, but has all (or at least the vast majority of) the commands from those packages.

Sections and Title

In order for the document to be correctly tagged, you need to set up sections and a title using their special commands. Sections, subsections, and subsubsections can be declared using the commands below. Subsections and subsubsections are not needed for the Canvas Ally checker, but sections are. It can also get upset if you have a subsection right after a section with no text in between.

```
\section{}           % This creates a section with a number.
\section*{}          % This creates a section without a number.
\subsection{}        % This creates a subsection with a number.
\subsection*{}        % This creates a subsection without a number.
\subsubsection{}     % This creates a subsubsection with a number.
\subsubsection*{}    % This creates a subsubsection without a number.
```

In addition, the title should be created with the `\title{}` command before `\begin{document}`. Then you can put the `\maketitle` command after `\begin{document}` like so:

```
\title{\textbf{Making Accessible PDFs with \LaTeX}} % Defines the title
\begin{document}
\maketitle % Creates the title
\section*{\underline{Initial Setup}} % Creates a numberless section
\end{document}
```

Putting text modifiers like `\large`, `\textbf`, `\underline`, or even `\vspace` in the `\title{}` or `\section{}` commands works fine without affecting accessibility. Mathematical expressions can be put in there as well.

Incompatible Packages

There are some packages that are incompatible with LaTeX automatic tagging. These can be found on this webpage: <https://latex3.github.io/tagging-project/tagging-status/>. Many times, the document will compile without error if you are using one or more of these unsupported packages, but then the Canvas Ally accessibility checker will mark the documents as having some problems (it took me a long time to discover this).

Mathematics

Math equations should automatically pass the Ally accessibility checker on Canvas when generated with the Document Metadata given on Page 1, although I doubt any screen reader could actually effectively read the math in the PDFs. For PDFs version 2.0, which, again, is not currently supported by the Ally Canvas checker, the MathML can be embedded directly in the PDF so a screen reader could, theoretically, read the math.

The following mathematical expressions have no problem passing the checker whether they are in the `equation` or `align` environments, in `$$ $$` or `\[ \]`, or inline using `$ $` or `\( \)`:

Bayes' Theorem can be written: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)}$.

$$\mathbf{X} = \begin{bmatrix} 1 & 12 & -2 \\ -2 & 1 & 0 \\ 4 & 1 & 2 \end{bmatrix} \quad (1)$$

$$\begin{aligned} MSE(\alpha) &= \sigma^2 \sum_{i=1}^r \left(\frac{\sigma_i^2}{\sigma_i^2 + \alpha} \right)^2 \frac{1}{\sigma_i^2} + \sum_{i=1}^r \left(\frac{\sigma_i^2}{\sigma_i^2 + \alpha} - 1 \right)^2 (\mathbf{v}_i^T \mathbf{x})^2 \\ &= \sigma^2 \sum_{i=1}^r \frac{\sigma_i^2}{(\sigma_i^2 + \alpha)^2} + \sum_{i=1}^r \left(\frac{-\alpha}{\sigma_i^2 + \alpha} \right)^2 (\mathbf{v}_i^T \mathbf{x})^2 \\ &= \sum_{i=1}^r \frac{\sigma^2 \sigma_i^2 + \alpha^2 (\mathbf{v}_i^T \mathbf{x})^2}{(\sigma_i^2 + \alpha)^2} \end{aligned}$$

Figures and Alternative Text

When including an image, it usually needs to have alternative text provided that describes what the image is showing. This can be added using the `alt={}` option in `\includegraphics` like this:

```
\includegraphics[alt={Include alt text here!}]{image.ext}
```

As an example, we can include the image on the top of the next page using this code:

```
\includegraphics[
width=2in,
alt={Image of the SUU campus showing the bell tower in the center with Braithwaite
Liberal Arts Center on the left and Old Main on the right.}
]{suu.jpg}
```

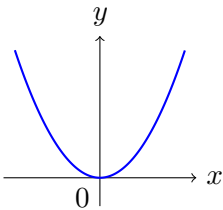


This also works with `tikz` and `tikzpicture`. When setting up `tikzpicture`, you can add the alt text using

```
\begin{tikzpicture}[alt={Include alt text here!}]
  tikz code here
\end{tikzpicture}
```

For instance, you could do something like this:

```
\begin{tikzpicture}[scale=0.75, alt={Graph of y equals x squared.}]
  \draw[->] (-1.7,0) -- (1.7,0) node[right] {$x$};
  \draw[->] (0,-0.5) -- (0,2.5) node[above] {$y$};
  \draw[blue, thick, domain=-1.5:1.5, samples=100] plot (\x, {\x*\x});
  \node[below left] at (0,0) {$0$};
\end{tikzpicture}
```

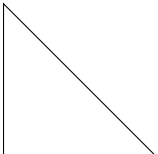


In `tikz`, you can type

```
\tikz[alt={Include alt text here!}] tikz code here
```

Here is an example:

```
\tikz[alt={Image of a right triangle.}] \draw (0,0) -- (2,0) -- (1,2) -- cycle;
```



You can also **mark the image as decorative** using the `artifact` option. These images do not need alt text.

```
\includegraphics[artifact]{image.ext}
```

An example for this:

```
\includegraphics[width=2in, artifact]{suu_logo.jpg}
```



It should be noted that the Ally accessibility checker does not know if the alternative text provided is actually helpful. For example, you could put `alt={I'm sick of alt text.}` and that will satisfy the checker even though that is not all that useful for a student.

Color Contrast

We do want to be mindful of the **color contrasts** when creating tikz pictures or including images created in other applications like Excel with colored cells. The file may be flagged if the contrast between colors is too low. The contrast should be at least 4.5:1.

There are online color contrast checkers like this one here: <https://webaim.org/resources/contrastchecker/> or you can install an application like this one: <https://vispero.com/lp/color-contrast-checker/> that has an eye drop tool so you can click directly on the text and background colors to check their contrast. This tool is more strict at marking something compliant than Ally, but gives the contrast number, which, again, should be at least 4.5:1 for Ally.

Excel also has an accessibility checker built in that checks for color contrast. You can find instructions for how to use it at [this link](#).

Tables

For each table, you will need to tell LaTeX where the headings are using a command like

`\tagpdfsetup{table/header-rows={1}}` if the heading is in the first row

or

`\tagpdfsetup{table/header-columns={1}}` if the heading is in the first column

or

`\tagpdfsetup{table/tagging=presentation}` if the table has no heading.

For this table below, for example, it would be set up like this because the first column has information about the content in the table.

```
\begin{table}
  \tagpdfsetup{table/header-columns={1}}
  \begin{tabular}{|c|c|c|c|c|}
    \hline
    Grade & A & B & C & D & F \\
    \hline
    \# of students & 18 & 30 & 18 & 7 & 4 \\
    \hline
  \end{tabular}
\end{table}
```

Grade	A	B	C	D	F
# of students	18	30	18	7	4

whereas this next table would have this following setup since the first row has the table heading and the second row has the column titles. So, we should specify both are header rows.

```
\tagpdfsetup{table/header-rows={1,2}} % No need to have \begin{table}. Put before tabular.
\begin{tabular}{|c|c|c|c|c|}
  \hline
  \multicolumn{5}{|c|}{\textbf{Five-Number Summary}} \\
  \hline
  Min & $Q_1$ & Median & $Q_3$ & Max \\
  \hline
  80.00 & 85.75 & 92.50 & 106.25 & 111.00 \\
  \hline
\end{tabular}
```

Five-Number Summary				
Min	Q_1	Median	Q_3	Max
80.00	85.75	92.50	106.25	111.00